In [84]:
```
import pandas as pd
import numpy as np
import seaborn as sns
import matplotlib.pyplot as plt
from sklearn.tree import DecisionTreeClassifier
from sklearn.metrics import confusion_matrix
from sklearn.model_selection import train_test_split
```

In [85]:
```
df = pd.read_csv("weather-check.csv")
df.head()
```

Out[85]:

	RespondentID	Check_Weather_Daily	Check_Weather_Method	Weather_Smartwatch_Likelihood	Age	Gender	Household_Income	US Regio
0	3887201482	Yes	The default weather app on your phone	Very likely	30 - 44	Male	$50,000 to 74,999	South Atlanti
1	3887159451	Yes	The default weather app on your phone	Very likely	18 - 29	Male	Prefer not to answer	South Atlanti
2	3887152228	Yes	The default weather app on your phone	Very likely	30 - 44	Male	$100,000 to 124,999	Middl Atlanti
3	3887145426	Yes	The default weather app on your phone	Somewhat likely	30 - 44	Male	Prefer not to answer	South Atlanti
4	3887021873	Yes	A specific website or app (please provide the ...	Very likely	30 - 44	Male	$150,000 to 174,999	Middl Atlanti

In [86]:
```
df.isnull().sum()
```

Out[86]:
```
RespondentID                    0
Check_Weather_Daily             0
Check_Weather_Method            0
Weather_Smartwatch_Likelihood   0
Age                             0
Gender                          0
Household_Income                0
US Region                       0
dtype: int64
```

In [87]:
```
df.duplicated().sum()
```

Out[87]:
```
np.int64(0)
```

In [88]:
```
df['Weather_Smartwatch_Likelihood'].value_counts()
```

Out[88]:
```
Weather_Smartwatch_Likelihood
Very likely          362
Somewhat likely      275
Very unlikely        209
Somewhat unlikely     73
Name: count, dtype: int64
```

In [89]:
```
df.columns
```

Out[89]:
```
Index(['RespondentID', 'Check_Weather_Daily', 'Check_Weather_Method',
       'Weather_Smartwatch_Likelihood', 'Age', 'Gender', 'Household_Income',
       'US Region'],
      dtype='object')
```

In [90]:
```
df.info()
```

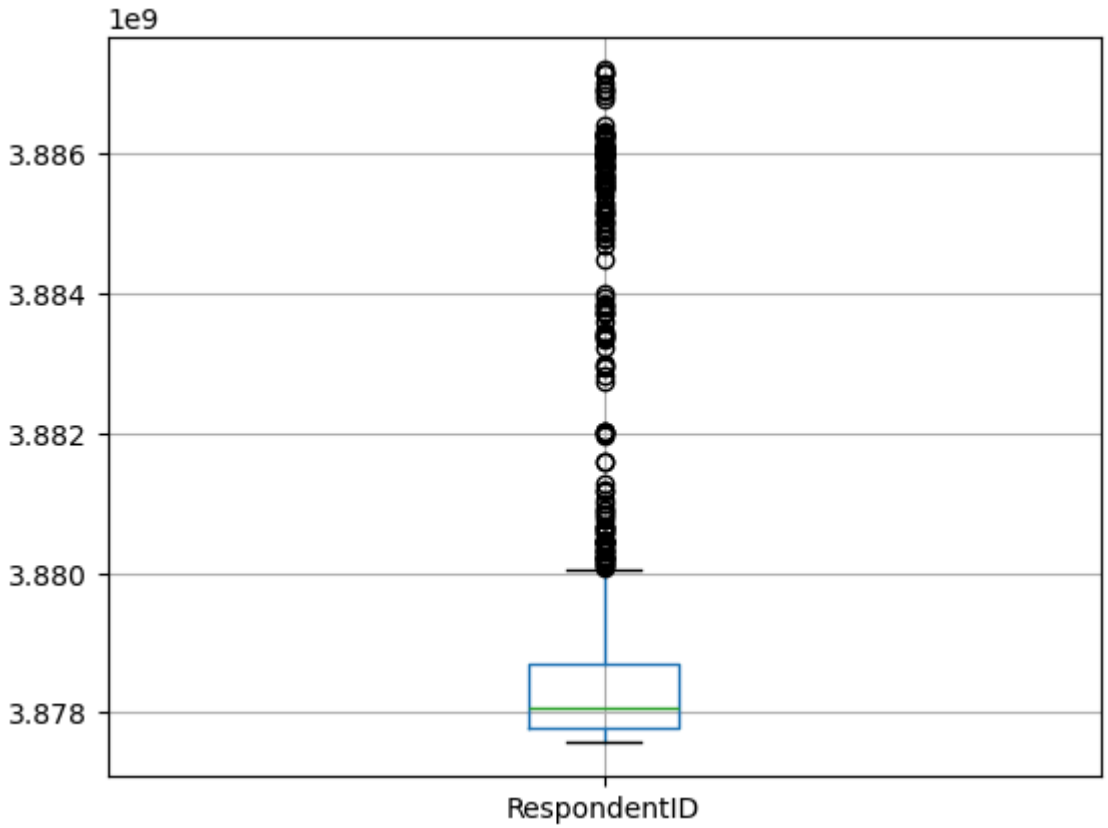
```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 919 entries, 0 to 918
Data columns (total 8 columns):
 #   Column                          Non-Null Count  Dtype
---  ------                          --------------  -----
 0   RespondentID                    919 non-null    int64
 1   Check_Weather_Daily             919 non-null    object
 2   Check_Weather_Method            919 non-null    object
 3   Weather_Smartwatch_Likelihood   919 non-null    object
 4   Age                             919 non-null    object
 5   Gender                          919 non-null    object
 6   Household_Income                919 non-null    object
 7   US Region                       919 non-null    object
dtypes: int64(1), object(7)
memory usage: 57.6+ KB
```

In [91]:
```
df.describe()
```

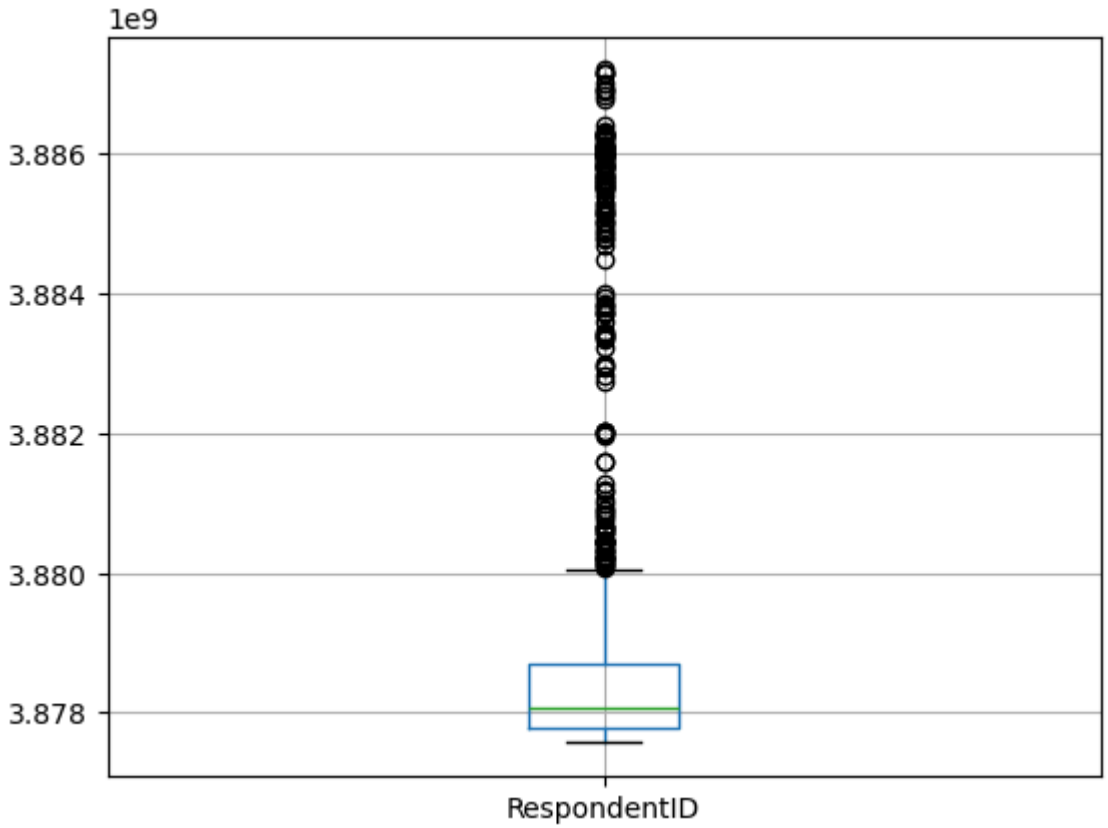
Out[91]:

	RespondentID
count	9.190000e+02
mean	3.879084e+09
std	2.445434e+06
min	3.877567e+09
25%	3.877793e+09
50%	3.878059e+09
75%	3.878700e+09
max	3.887201e+09

In [92]:
```
df.select_dtypes(include='number').boxplot()
plt.show()
```



In [93]:
```
df.boxplot(column='RespondentID')
plt.show()
```



In [94]:
```
model = DecisionTreeClassifier()
```

In [95]:
```
X = df.drop('Weather_Smartwatch_Likelihood', axis=1)
y = df['Weather_Smartwatch_Likelihood']
```

In [96]:
```
X = pd.get_dummies(X, drop_first=True)
```

In [97]:
```
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2, random_state=42)
```

In [98]:
```
model = DecisionTreeClassifier(random_state=42)
model.fit(X_train, y_train)
```

Out[98]:

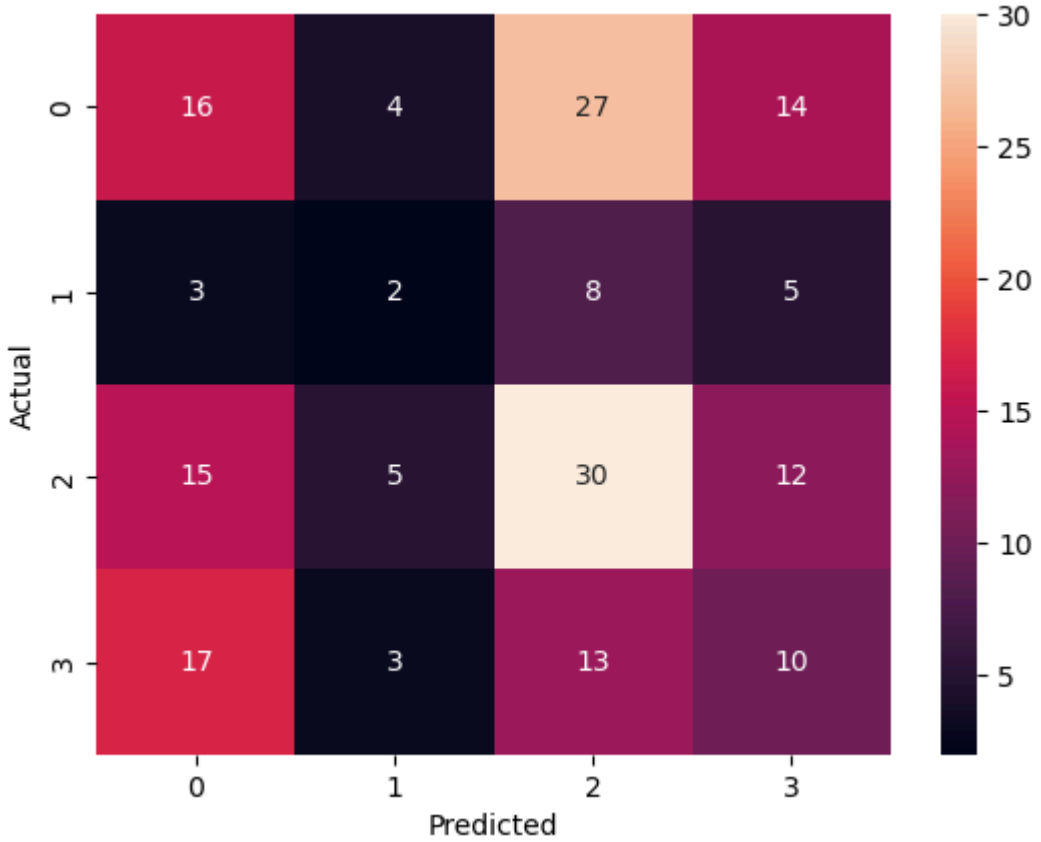
▾ DecisionTreeClassifier

▶ Parameters

In [99]:
```
y_pred = model.predict(X_test)
```

In [100…
```
cm = confusion_matrix(y_test, y_pred)
```

In [101…
```
sns.heatmap(cm, annot=True, fmt='d')
plt.xlabel('Predicted')
plt.ylabel('Actual')
plt.show()
```


16
4
27
14
3
2
8
5
15
5
30
12
17
3
13
10
0
1
2
3
Actual
Predicted
30
25
20
15
10
5